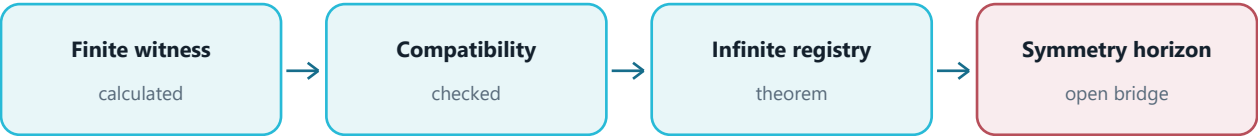


From Finite Witness to Infinite Registry

Budgeted residue construction, p-adic forests, TUFT transformation registries, and symmetry horizons

A proof-bearing atlas that keeps finite computation, theorem-governed continuation, canonical input, and open construction obligations visibly distinct.



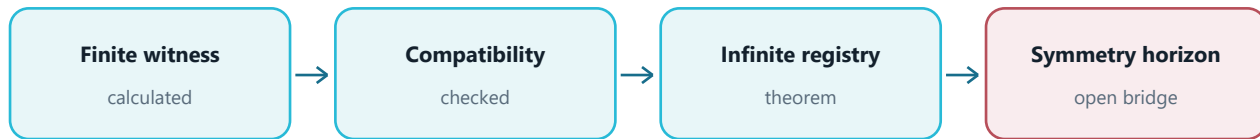
The seam is visible: calculation ends; theorem governs continuation; open obligations remain open.

CALCULATED	THEOREM-RENDERED	OPEN
Budgeted prime witnesses, cyclic orbits, CRT, Möbius and sl_2 certificates, J coefficients.	p-adic/profinite limits, Lie realization route, established moonshine theorem.	TUFT-to-moonshine construction; any separate physical or stronger-logical extension.

Laboratory notice. Standard mathematics and cited results remain attributed to their established sources. No claim of discovery, invention, ownership, priority, or prior art is made for established mathematics or methods. The contribution claimed is the declared registry synthesis, exact claim discipline, canonical integration, and executable atlas.

Version 1.0 · 26 July 2026 · Companion: finite_witness_infinite_registry_atlas.html

Abstract



The seam is visible: calculation ends; theorem governs continuation; open obligations remain open.

We present a finite-witness protocol for studying unbounded mathematical registries without treating infinity as a computational endpoint. A run is defined by a frame, an operation budget, a witness contract, and an explicit stopping state. Completed finite certificates are separated from unresolved candidates, theorem-governed continuations, imported canonical data, and open conjectural bridges.

The protocol is instantiated in five linked modules. First, two sequential prime classifiers are compared under an equal logical-token ledger. Testing only already certified primes has no greater per-candidate divisibility cost than testing all integer divisors, hence reaches a certified horizon at least as large under every common budget; reproducible runs through five million tokens quantify the strict advantage for the selected budgets. Second, linear and cyclic descriptions are related through an explicit quotient seam, yielding the orbit identity $|k|=n/\gcd(n,k)$ and its prime criterion. Third, finite residue trees are placed inside the standard inverse systems defining Z_p and $Z\text{-hat}$; the browser enumerates finite prefixes and theorem-renders, rather than simulates, the unbounded continuation. Fourth, the TUFT registry discipline is made exact through Möbius certificates and the smooth left-translation realization of finite-dimensional Lie groups, extended to finite-dimensional real Lie algebras by Lie's third theorem. Fifth, exact coefficients of $J(q)=j(q)-744$ are calculated and compared with canonical Monster representation dimensions, while the proposed TUFT-to-moonshine connection is retained as an open construction obligation.

The result is a reproducible proof demonstrator for a modest but substantial thesis: finite, frame-declared witnesses can support rigorous traversal into mathematically infinite registries when every transition states what was calculated, what follows from theorem, what was imported, and what remains unproved. No physical law, universal $1/f$ claim, logical independence result, or derivation of monstrous moonshine is asserted.

Keywords: finite witness; operation budget; prime certificate; p-adic integers; profinite completion; registry; Möbius transformation; Lie group; inverse limit; monstrous moonshine; proof demonstrator.

1. Contribution and claim classes

The central methodological error this work avoids is treating a finite rendering as though it had exhausted the object it depicts. A browser cannot complete an infinite sieve, enumerate Z_p , list an infinite product, traverse every diffeomorphism of a Lie group, or reproduce Borchers' proof by drawing a suggestive graph. It can, however, calculate finite witnesses and place them under exact continuation theorems.

We therefore use six visible status classes.

Status	Meaning
Framework definition	A declared protocol term; valid only inside its stated frame.
Proved internal	Derived here from declared premises and definitions.
Established standard	External mathematics used with attribution; not claimed as original.
Calculated	Recomputed by the browser or reference program with finite arithmetic.
Theorem-rendered	Shown graphically from a theorem without exhaustive enumeration.
Canonical input	Established data imported with attribution.

Status	Meaning
Open	A proposed extension with an unsatisfied proof or construction obligation.

These labels are epistemic types, not decoration. Substituting one for another is a category error in the demonstrator.

1.1 What “beyond” means here

The phrase finite witness to infinite registry does not mean that a program reaches infinity. Nor does “ZFC and beyond” mean that this paper proves independence from ZFC, exceeds its consistency strength, or constructs a stronger foundational theory. All standard objects used here can be formalized in ordinary set-theoretic mathematics. Here beyond means beyond the finite calculated prefix into an unbounded object whose continuation is specified by theorem. Any future IRRPROOF kernel that asserts a genuinely stronger logical claim must state its axioms, semantics, proof rules, soundness target, and relative-consistency obligations separately.

2. Frames, witnesses, residue, and budgets

Definition 2.1 (frame)

A frame is a tuple

$$F=(S,D,R,W,T,C),$$

where S is a carrier of admissible states, D a family of admitted distinctions, R a family of relations, W a witness contract, T a family of transformations, and C a collection of stopping conditions. This refines the framing discipline of The Geometry of Framing: Rigorous Canonical.

Definition 2.2 (witness state)

A witness state is a record

$$\text{omega}=(F,x,P,\text{tau},\text{sigma}),$$

where x is the object examined, P the property tested, tau an audit trace, and sigma in certified,refuted,unresolved. A witness is frame-relative: it establishes exactly what W licenses inside F, not an unspecified universal truth.

Definition 2.3 (translation residue)

Let $\phi:F \rightarrow F'$ translate a state or certificate between frames. The translation residue is the set of declared features of the source whose preservation has not been proved under ϕ . It is not automatically error; it is unverified transport.

Definition 2.4 (budgeted run)

A budgeted run is

$$R=(F,A,x_0,B,\text{ledger}),$$

where A is an algorithm, x_0 a starting state, B in N a spend limit, and ledger an operation ledger assigning cost to admissible operations. The output contains all completed witnesses, the exact spend, and any unresolved frontier state.

Proposition 2.5 (sound budget termination)

If a certificate requires a finite set of checks $Q(x)$, then a run stopped before every required check has either succeeded or a refuting witness has been found must label x unresolved.

Proof. By the witness contract, certification requires completion of $Q(x)$ unless a refutation condition has already terminated the test. At budget exhaustion neither condition holds. Calling x certified or refuted would assert a premise absent from the audit trace. Therefore the only sound status is unresolved. $\square$

This elementary proposition is operationally important. The unresolved frontier is not discarded noise: it is the exact seam between what the run witnessed and what it did not.

3. Primitive registry ladder

The primitive ladder supplies a dependency vocabulary, not a uniqueness theorem.

1. A mark introduces one distinguishable referent.
2. A degree of freedom permits an admissible change.
3. A trace records an ordered sequence of changed states.
4. A return relation identifies a later state with an earlier state.
5. A seam records where that identification was imposed.
6. A branch records multiple admissible continuations.
7. A grid combines compatible distinctions or traces into a comparative registry.

The distinction between linear and chiral/cyclic presentation becomes precise at step 4: closure is not silently present in a line. It is an additional relation. This is enough to compare unwrapped enumeration with residue classes, and binary branching with general p-branching, without claiming that these objects have been logically derived from phenomenological language alone.

4. A budgeted prime-witness theorem and experiment

4.1 Two declared probes

Both algorithms inspect candidates $n=2,3,4,\dots$ in order.

- All-divisor probe A: test $d=2,3,\dots, n$ until $d|n$ or the list is exhausted.
- Certified-prime probe P: test only previously certified primes $p \leq \sqrt{n}$ until $p|n$ or the list is exhausted.

The ledger charges one token for starting a candidate and one token for each divisibility test. Certificate writes and elapsed milliseconds are recorded but are not part of the equal-budget currency.

Lemma 4.1 (prime-divisor sufficiency)

For $n > 1$, if n is composite then it has a prime divisor $p \leq \sqrt{n}$. Consequently, failure of every prime-divisor test through $\sqrt{n}$ is a prime certificate.

Proof. Write $n=ab$ with $1 < a \leq b$. Then $a \leq \sqrt{n}$, and some prime $p|a$ satisfies $p \leq a \leq \sqrt{n}$. $\square$

Lemma 4.2 (per-candidate dominance)

For every candidate n , after all smaller candidates have been correctly classified, probe P performs no more divisibility tests than probe A.

Proof. If n is prime, A tests every integer from 2 through n , whereas P tests only the primes in that interval. If n is composite, let p be its least prime divisor. No integer $d < p$ divides n . Probe A therefore tests all integers from 2 through p ; probe P tests only primes through p . In either case the test list of P is a subsequence of the relevant list of A. $\square$

Theorem 4.3 (equal-ledger certified-horizon dominance)

Let $H_P(B)$ and $H_A(B)$ be the largest completely classified candidates under budget B . Under the declared sequential order and token ledger,

$$H_P(B) \geq H_A(B) \text{ for every } B \text{ in } \mathbb{N}.$$

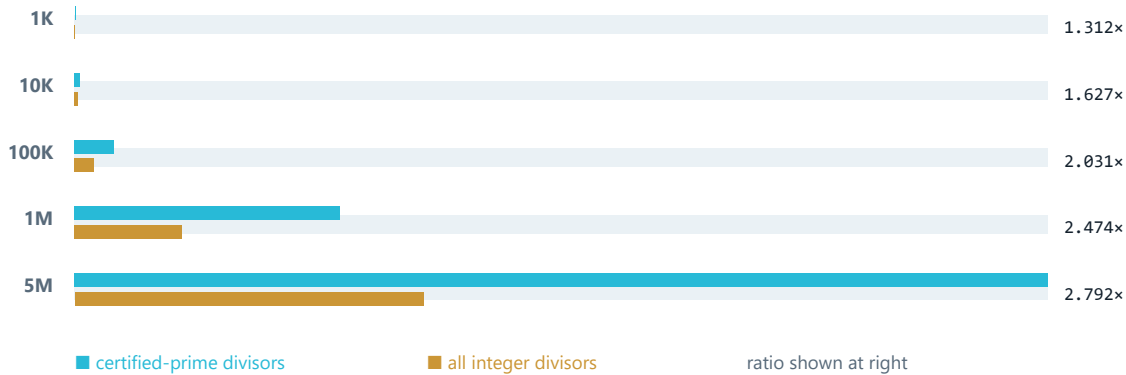
Proof. Both probes pay the same one-token candidate-start cost. By Lemma 4.2, the cumulative cost through every completed horizon h is no greater for P than for A . Hence any horizon affordable to A is affordable to P . Budget exhaustion within a candidate is handled by Proposition 2.5. $\square$

Scope note

Theorem 4.3 compares exactly these two ordered trial-division probes under exactly this logical ledger. It does not claim superiority over the Sieve of Eratosthenes, segmented sieves, Pritchard-family linear sieves, probabilistic primality testing, AKS, or specialized large-prime software. The prime-only restriction is standard mathematics; no claim of its invention is made.

4.2 Reproducible results

Certified horizon under identical logical-token budgets



Budget	H_P	H_A	Gain	Horizon ratio	primes certified by P
1,000	307	234	73	1.3120	63
10,000	2,339	1,438	901	1.6266	346
100,000	16,916	8,328	8,588	2.0312	1,950
1,000,000	114,912	46,440	68,472	2.4744	10,866
5,000,000	421,356	150,910	270,446	2.7921	35,492

The reference program and independent Python implementation agree at all ten method–budget combinations. Independent Eratosthenes counts equal the reported number of primes at every certified horizon. The increasing empirical ratio is descriptive of these runs; no asymptotic law is inferred from five budgets.

4.3 What the experiment demonstrates

The experiment reconstructs a familiar arithmetic fact in witness-bearing form: a structured divisor registry avoids logically redundant composite divisors. Its significance for the larger architecture is not a new prime algorithm. It is the clean separation of (i) admissible probes, (ii) finite certificates, (iii) spend, and (iv) unresolved continuation. The same discipline can govern more elaborate IRRPROOF search kernels.

5. Line and cycle as different frames

Let the linear frame be $(\mathbb{Z}, +)$. Imposing $x \sim y$ exactly when $x - y \in n\mathbb{Z}$ produces the quotient $\mathbb{Z}/n\mathbb{Z}$. The quotient map preserves addition but not absolute linear position. The lost integer lift is translation residue unless a representative convention is declared.

Proposition 5.1 (additive orbit length)

For k in $\mathbb{Z}/n\mathbb{Z}$, the orbit generated by addition of k has length

$$\text{ord}_+(k) = (n)/(\gcd(n,k)).$$

Proof. The first return occurs at the least $m > 0$ for which $mk \equiv 0 \pmod n$, equivalently $n \mid mk$. Writing $n = gn'$, $k = gk'$, with $g = \gcd(n,k)$ and $\gcd(n',k') = 1$, the least such m is n' . $\square$

Corollary 5.2 (additive prime criterion)

For $n > 1$, n is prime if and only if every nonzero k in $\mathbb{Z}/n\mathbb{Z}$ generates a full orbit.

Proof. Every nonzero $k < n$ is coprime to n exactly when n has no nontrivial divisor. Apply Proposition 5.1. $\square$

This criterion is not a fast primality test: checking all steps would merely restate divisibility. Its value here is representational. It shows how primality can appear as absence of orbit fragmentation after a declared wrap.

6. From residue trees to p-adic and profinite registries

Proposition 6.1 (exact lift count)

Every class $a \pmod{p^k}$ has exactly p lifts modulo p^{k+1} :

$$a + t p^k \pmod{p^{k+1}}, \quad t = 0, 1, \dots, p-1.$$

Proof. The displayed classes all reduce to $a \pmod{p^k}$. Two are congruent modulo p^{k+1} only when $p \mid (t-t')$, which within the stated range forces $t=t'$. Every lift has one of these residues. $\square$

Thus the 2-adic residue tree literally bifurcates, while the general p -adic tree has p children at every node. At level k there are exactly p^k residue nodes. A browser may enumerate those nodes only while p^k lies within its declared node cap. It may always calculate the count using integer exponentiation.

Definition 6.2 (p-adic inverse limit)

With reduction maps $\rho_{k+1,k}: \mathbb{Z}/p^{k+1}\mathbb{Z} \rightarrow \mathbb{Z}/p^k\mathbb{Z}$,

$$\mathbb{Z}_p = \lim_{\leftarrow} \mathbb{Z}/p^k\mathbb{Z}$$

is the set of compatible sequences (x_k) satisfying $\rho_{k+1,k}(x_{k+1}) = x_k$. The infinite object is not the last level of a large finite tree. It is the compatibility object over all levels.

Theorem 6.3 (profinite completion of the integers)

Standard profinite number theory and the Chinese remainder theorem give

$$\widehat{\mathbb{Z}} = \lim_{\leftarrow} \mathbb{Z}/n\mathbb{Z} \cong \prod_{p \text{ prime}} \mathbb{Z}_p.$$

The atlas calculates finite CRT witnesses and finite prime/depth rectangles. It theorem-renders the inverse limit and full product. No finite rectangle is labeled as the completed profinite object.

6.1 Registry interpretation

The p -adic forest supplies an exact instance of finite witnesses cohering into an infinite registry:

- nodes are finite residue states;
- edges are lift relations;

- reduction maps test compatibility;
- finite prefixes are calculated;
- the inverse limit is established by definition and theorem;
- omitted individual nodes remain outside the finite witness, not falsely summarized as computed.

This is a stronger and cleaner result than an appeal to visual resemblance. The branching law, compatibility maps, and limiting object are all specified.

7. TUFT as a transformation-registry discipline

Definition 7.1 (registry)

A registry is a tuple

$$R=(S,O,M,I),$$

where S is a carrier, O a declared observable family, M admissible transformations or operations, and I invariants or compatibility conditions. A registry morphism must state which components and invariants it preserves. Undeclared resemblance is not a morphism.

This definition follows the Registry Factorization Principle. The proposed full registry category, functorial semantics, and gluing theorem remain open where their composition laws and universal properties have not been supplied.

7.2 Möbius maps as an exact TUFT checkpoint

For

$$M=[[a, b]; [c, d]], M! = o,$$

the projective class $[M]$ acts by $z(az+b)/(cz+d)$. Nonzero scalar multiples represent the same map. The quantity

$$\chi(M)=(\text{tr}(M)^2)/(\det M)$$

is scale invariant because numerator and denominator both acquire a factor λ^2 . Fixed points satisfy

$$cz^2+(d-a)z-b=0,$$

with discriminant

$$\Delta=(a+d)^2-4(ad-bc)=\text{tr}(M)^2-4 \det M.$$

For real coefficients, the sign of Δ distinguishes the familiar two-real, repeated-real, and conjugate-pair fixed-point cases. The atlas calculates this certificate directly. The standard three-point theorem—three distinct source points and three distinct target points determine a unique Möbius map—is theorem-rendered as a proof-transport pattern.

This checkpoint matters because carrier, quotient, invariant, and transport are explicit. It is therefore a genuine registry realization, not a coincidence of plotted coordinates.

8. Smooth registry realizations of Lie groups and algebras

Theorem 8.1 (left-translation realization)

Every finite-dimensional Lie group G admits a faithful smooth registry realization

$$L:G \rightarrow \text{Diff}(G), g \mapsto L_g, \text{quad } L_g(x)=gx.$$

Proof. Each L_g is a diffeomorphism with inverse L_g^{-1} . The map is a homomorphism because $L_g L_h = L_{gh}$. It is faithful: if $L_g = L_h$, evaluation at the identity gives $g = h$. Smoothness follows from smooth group multiplication. $\square$

Theorem 8.2 (infinitesimal recovery)

The tangent space $T_e G$ identifies with the left-invariant vector fields. Under this identification the Lie bracket is the commutator of infinitesimal registry transports.

This is standard Lie theory and is also proved in Supplement A: Smooth Registry Realizations of Lie Groups. The atlas checks the exact $\mathfrak{sl}_2$ witness

$$H = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, E = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, F = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix},$$

for which

$$[H, E] = 2E, [H, F] = -2F, [E, F] = H,$$

and the Jacobi sum is the zero matrix.

Corollary 8.3 (finite-dimensional real Lie algebras)

Every finite-dimensional real Lie algebra $\mathfrak{g}$ integrates to a connected simply connected Lie group G by Lie's third theorem, and therefore receives the smooth registry realization of Theorem 8.1 with infinitesimal recovery by Theorem 8.2.

Scope boundary

Corollary 8.3 is broad, but not unlimited. It does not cover arbitrary infinite-dimensional Lie algebras, topological Lie algebras, Lie superalgebras, vertex algebras, or generalized Kac–Moody algebras without additional hypotheses and constructions. In particular, it does not by itself construct the algebraic objects used in monstrous moonshine.

For compact connected G , a maximal torus T , Weyl group $W = N(T)/T$, and root-space decomposition supply a finer registry hierarchy. This is established Lie theory. The hierarchy can be represented without claiming exhaustive enumeration of G .

9. The moonshine horizon

9.1 Exact modular calculation

Let

$$E_4(q) = 1 + 240 \sum_{n \geq 1} \sigma_3(n) q^n, E_6(q) = 1 - 504 \sum_{n \geq 1} \sigma_5(n) q^n,$$

and

$$\Delta(q) = (E_4(q)^3 - E_6(q)^2) / (1728), j(q) = (E_4(q)^3) / (\Delta(q)), J(q) = j(q) - 744.$$

Finite BigInt power-series arithmetic in the reference program gives

$$J(q) = q^{-1} + 196884q + 21493760q^2 + 864299970q^3 + 20245856256q^4 + \dots$$

These coefficients are calculated from the displayed formulas; no floating-point approximation is used.

9.2 Canonical imported calibration

Two famous comparisons are

$$196884 = 1 + 196883,$$

$$21493760 = 1 + 196883 + 21296876,$$

where the terms on the right are canonical Monster representation dimensions. The dimensions are imported data. Their appearance is not derived by the prime probe, the p-adic tree, or the finite-dimensional TUFT realization.

The Conway–Norton moonshine conjectures and Borchers' theorem concern graded Monster modules, trace functions, modular functions, and generalized Kac–Moody structure. That established theory is attributed to its sources.

Open Obligation 9.1 (TUFT-to-moonshine bridge)

To promote the suggested structural relationship to a theorem, one must construct at least:

- 1. a specified graded registry object $V=\text{direct_sum}_{(n \geq -1)} V_n$;
- 2. a Monster action, or an explicitly equivalent automorphism construction, on V ;
- 3. graded trace functions $T_g(q)=\text{sum_ntr}(g|V_n)q^n$;
- 4. proofs of the required modularity, genus-zero, or replicability properties;
- 5. compatibility maps showing that the TUFT observables and transformations recover—not merely resemble—the established moonshine data.

Until these are supplied, the bridge is open. This is not dismissal: it is a precise research program.

10. Calculated, theorem-rendered, and imported layers

Module	Calculated in browser	Theorem-rendered	Imported / open boundary
Prime probe	candidate starts, divisor tests, witnesses, horizon, frontier	general horizon-dominance proof displayed	no universal algorithmic superiority claimed
Line/cycle	selected gcd and orbit	orbit formula and prime criterion	quotient seam must be declared
p-adic	finite level counts and displayed nodes	inverse limit and unbounded lift continuation	individual omitted nodes not computed
Profinite	finite CRT witness	$Z\text{-hat}\approx\text{product}_p Z_p$	full product not enumerated
Möbius/TUFT	determinant, trace, chi, discriminant	three-point theorem and projective transport	broader gluing framework open
Lie	exact sl_2 matrices	left translations, infinitesimal recovery, Lie III corollary	infinite-dimensional extension not asserted
Moonshine	initial J-coefficients	established theorem summarized	Monster dimensions imported; TUFT bridge open

This table is the paper's central checksum. Any future module should be rejected if it cannot identify its row entries.

11. Relation to IRRPROOF and residue-aware computation

The demonstrator suggests a kernel contract for residue-aware computation.

Definition 11.1 (residue-aware kernel record)

For each transition $s_i \rightarrow s_{i+1}$, record

$$K_i=(F_i,s_i,o_i,w_i,I_i,r_i,B_i),$$

where F_i is the active frame, o_i the operation, w_i its witness or failure state, I_i the invariants checked, r_i the unverified translation residue, and B_i the remaining budget.

Such a record supports replay, comparison, branching, and adversarial audit. A frozen archive is a state description of a run, not the run's living temporal continuum. Its mathematical value lies in the exact transition log and stopping seam, not in pretending to capture every possible continuation.

For IRRPROOF, extensions can be packaged as kernels or macros above the base engine only when they declare:

- input and output registries;
- admitted inference rules;
- preserved invariants;
- witness and stopping conditions;
- resource ledger;
- unresolved residue;
- external theorem dependencies;
- proof obligations not discharged by the engine.

This makes the architecture extensible without making it authoritative. A module can propose a path, supply evidence, and invite challenge while leaving interpretation and adoption to the observer or user.

12. Adversarial tests and falsification conditions

The work invites attack at exact joints.

Target	A successful challenge would show
Budget soundness	a candidate labeled certified despite an unperformed required check
Horizon dominance	a budget under the declared ledger for which $H_P(B) < H_A(B)$, or a flaw in Lemma 4.2
Reproducibility	disagreement among browser, Node reference, and independent Python results after matching the ledger
Cycle module	an orbit whose measured return differs from $n/\gcd(n,k)$
p-adic module	a residue with other than p distinct lifts, or a displayed edge violating reduction compatibility
Möbius module	failure of chi under nonzero scalar rescaling, outside a declared degenerate case
Lie route	a failure of faithfulness of left translation, or a scope claim exceeding finite-dimensional hypotheses
Moonshine bridge	currently not falsified because it is not asserted; completion requires all obligations in §9.2

The protocol is intentionally assailable. The goal is not to make challenge impossible, but to make every challenge land on a statement whose premises and status are visible.

13. Limitations

1. The prime experiment compares two elementary classifiers, not state-of-the-art prime generation.
2. Logical tokens are a declared abstract cost model. They are reproducible but not identical to CPU instructions, energy, or elapsed time.
3. Browser timing varies by device and is not used to prove the mathematical comparison.
4. Finite p-adic drawings do not approximate every topological property merely by becoming large.
5. The TUFT label contributes an organizing registry discipline; it does not supersede standard projective or Lie theory.
6. The finite-dimensional Lie route does not construct generalized Kac–Moody or vertex-algebra data.
7. The moonshine comparison is a calibrated horizon, not a derived correspondence.
8. No claim about physical energy, spacetime singularities, biological agency, universal 1/f behavior, or continuum ontology follows from these mathematical modules.

9. Observer-relative framing does not imply that mathematical truth is arbitrary. Once a frame and morphism are declared, their internal consequences remain objectively contestable.

14. Conclusion

The strongest result is not that a browser has approached infinity. It is that the architecture refuses to confuse three different acts:

finite calculation != theorem-governed infinite construction != speculative structural extension.

Within that separation, a coherent ladder is obtained. Minimal distinctions support traces and explicit quotient seams. Divisibility produces finite prime and composite witnesses. Residue classes organize into exact p-branching inverse systems. Profinite structure combines the prime-indexed lanes. TUFT supplies a disciplined language for carriers, transformations, invariants, and transport. Möbius and finite-dimensional Lie realizations provide rigorous transformation checkpoints. Modular coefficients can be calculated at the edge of moonshine, where established canonical data begin and the proposed registry bridge remains visibly open.

This is enough to justify the demonstrator as a proof-bearing research instrument. It reconstructs its finite substrate, states the theorems governing its infinite registries, preserves the frontier that computation did not cross, and presents the next ambitious connection as an exact obligation rather than an accomplished fact.

Appendix A. Reference algorithm

```
procedure PROBE(B, DIVISORS)
  spent ← 0; n ← 2; primes ← []
  while spent < B
    spend START(n)
    for d in DIVISORS(n, primes) with  $d^2 \leq n$ 
      if spent = B
        return witnesses, UNRESOLVED(n, next=d)
      spend DIVISIBILITY_TEST(n,d)
      if d divides n
        write COMPOSITE(n,d); goto NEXT
    write PRIME(n)
    append n to primes
  NEXT:
    n ← n+1
  return witnesses, frontier n
```

The reference implementation also records method, operation counts, certificate counts, unresolved state, elapsed worker time, and—in the Node packet—a SHA-256 digest of the witness structure.

Appendix B. Reproducibility packet

The accompanying packet contains:

- the standalone executable atlas;
- this manuscript in Markdown and PDF;
- the full 23-item JSON claim ledger;
- the Node reference implementation and results;
- an independent Python validator and validation results;
- browser interaction tests and a manifest with SHA-256 hashes.

The atlas has no external libraries or network calls. It uses a Blob Web Worker for the prime experiment and BigInt arithmetic for modular coefficients. Desktop and mobile layouts are tested in Chromium.

Appendix C. Canonical source position

The following internal works supply terminology, earlier constructions, or rigorous checkpoints:

1. The Geometry of Framing: Rigorous Canonical — frame, witness, translation residue, and transport discipline.
2. Registry Factorization Principle: Formalized — registry tuple, morphism guardrails, and open categorical program.
3. Primitive Registry Ladder / Frame Imposition Synthesis v0.4 — active-provisional seed ladder and linear/chiral frame distinction.
4. TUFT Registry Studies: Structural Analysis for ZFC / IRRPROOF — Möbius structural certificates and laboratory notice.
5. Supplement A: Smooth Registry Realizations of Lie Groups — left-translation theorem, infinitesimal recovery, maximal-torus refinement.
6. Infinite Registry: Structured Map — distinction between a flat infinite collection and a compatibility-bearing structured registry.
7. R_S Protocol R11.23+ and the Rope-and-Sand series — historical provenance and motivation only; no rational-interpolation prime predictor is used here.

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End of independent laboratory record.